

Fig5 IMPG homolog-vs-interchromosomal class winners across the whole genome

Each row is a full-length CHM13 query chromosome. Colored tracts are 2 kb windows merged by target chromosome where best interchromosomal IMPG identity exceeds best same-chromosome/homolog identity.

Light context marks PAR1/PAR2, acrocentric PHR intervals, centromeres, and the chr9q-to-chr3q candidate windows. The plotted windows are from the corrected depth-filtered PAN027 paternal vs PAN011 class-winner summaries.

winning target chromosome

chr3

chr9

chr13

chr14

chr15

chr21

chr22

chrX

chrY

same/homolog wins or no retained comparison

candidate window outline

PAR / acrocentric / centromere context

SweepGA/FastGA mapping 1:1

18 kb interchromosomal-winner bp; 9 windows of 21 same+inter comparisons

largest interchromosomal-winner pairs

1:1 mapping

chr1 -> chr8

2 kb

chr11 -> chr9

2 kb

chr13 -> chr11

2 kb

chr17 -> chr8

2 kb

chr18 -> chr1

2 kb

4:4 mapping

chr13 -> chr21

442 kb

chrX -> chrY

350 kb

chr21 -> chr13

198 kb

chr21 -> chr14

124 kb

chr22 -> chr14

112 kb

10:10 mapping

chr13 -> chr21

858 kb

chr21 -> chr13

318 kb

chr22 -> chr15

238 kb

chr13 -> chr14

204 kb

chr21 -> chr14

188 kb

SweepGA/FastGA mapping 4:4

3.30 Mb interchromosomal-winner bp; 1,649 windows of 96,401 same+inter comparisons

SweepGA/FastGA mapping 10:10

3.56 Mb interchromosomal-winner bp; 1,778 windows of 174,942 same+inter comparisons

Candidate context

PAN028 chr9q->chr3q

chr9:134.38-134.88 Mb

PAN027 chr9q->chr3q

chr9:135.70-136.20 Mb

PAR1 X/Y

chrX:0.00-0.50 Mb

Controls

PAR1/PAR2 intervals are drawn on chrX/chrY.

Acrocentric PHR intervals are drawn on chr13/14/15/21/22.

0 50 100 150 200 native query chromosome coordinate (Mb)